Tongrui Zhang

PhD Student · College of Artificial Intelligence · Nankai University

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PROFILE

PhD student advised by Professor Jianda Han, studying how embodied systems can build reliable spatial understanding from incomplete and changing sensory observations. Current interests connect learning-based visual perception, localization, spatial representation, uncertainty-aware sensor fusion, map matching, and mobile-robot navigation. Earlier peer-reviewed work spans radar semantic segmentation and lithium-ion battery state estimation, fault diagnosis, and remaining-life prediction.

EDUCATION

Nankai University — PhD, Artificial Intelligence — Sep 2026–Present
College of Artificial Intelligence · Advisor: Professor Jianda Han · Tianjin, China

University of Calgary — MEng, Mechanical Engineering — Jun 2026
Energy and Environment specialization · Calgary, Canada · Degree formally awarded June 2026

Dalian Maritime University — BE — 2024
Mechanical Design, Manufacturing and Automation · Joint education program with the University of Houston

University of Houston — BS, Mechanical Engineering — 2024
Joint education program with Dalian Maritime University · Houston, United States

SELECTED RESEARCH INTERESTS

Learning-Based Visual Perception · Visual Localization · Spatial Representation · Uncertainty-Aware Multi-Sensor Fusion · Visual Navigation · Map Matching

PEER-REVIEWED PUBLICATIONS

[1] **Tongrui Zhang** and Hao Sun. “A Method for Predicting the Remaining Life of Lithium-Ion Batteries Based on an Improved Dempster–Shafer Evidence Theory Framework.” *Energies* 18(13), 3370, 2025. doi:10.3390/en18133370

[2] **Tongrui Zhang** and Hao Sun. “Research on Model Prediction of Remaining Service Life of Lithium-Ion Batteries Based on Chaotic Time Series.” *Electronics* 14(11), 2280, 2025. doi:10.3390/electronics14112280

[3] **Tongrui Zhang**, Ce Huang, and Qiujin Zhu. “Robust State of Charge Estimation for Lithium-Ion Batteries in Variable Temperatures Using Truncated Singular Value Decomposition Cubature Kalman Filter Algorithm.” *International Journal of Electrochemical Science* 19(5), 100515, 2024. doi:10.1016/j.ijoes.2024.100515

[4] **Tongrui Zhang** and Yunsheng Fan. “A 3D U-Net Based on a Vision Transformer for Radar Semantic Segmentation.” *Sensors* 23(24), 9630, 2023. doi:10.3390/s23249630

[5] **Tongrui Zhang**, Ran Li, and Yongqin Zhou. “Battery Fault Diagnosis Method Based on Online Least Squares Support Vector Machine.” *Energies* 16(21), 7273, 2023. doi:10.3390/en16217273

RESEARCH EXPERIENCE

Battery System Safety for Electric Vehicles — Team Member — Jan 2022–Nov 2023
Research project advised by Professor Xinping Qiu, Tsinghua University

- Curated battery performance and fault data for model development and validation.
- Implemented truncated-SVD cubature Kalman filtering for temperature-aware state-of-charge estimation.
- Developed an online least-squares support vector machine workflow for progressive fault diagnosis.

Amphibious Biomimetic Autonomous Underwater Vehicle — Team Member — May–Aug 2022
Dalian Maritime University

- Contributed to structural design, control-system development, and battery-system testing.
- Project received a first-class national award in the 1st World University Underwater Robots Competition.

Safety Guard: Vibration Measurement Analyzer — Team Member — Jan–Oct 2021
Nankai University

- Worked on MEMS-based vibration measurement, modular hardware, and FFT-based monitoring.
- Project received a national bronze award in the 7th China International College Students' Internet+ Innovation and Entrepreneurship Competition.

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ROBOTICS PROJECT

Remote Control and Target Recognition for a Dual-Screw Amphibious Robot 2024

Developed a Raspberry Pi–based control workflow and an improved YOLOv8s detector with model pruning and deployment on a Huawei Atlas 200I DK A2 edge platform. Against the YOLOv8s baseline, project evaluation reported gains of 6.9 percentage points in precision, 1.1 points in mAP@0.5, and 1.5 points in mAP@0.5:0.95 while maintaining recall.

Pruning reduced the reported parameter count from 11,126,358 to 8,936,846 and operations from 28.4G to 22.2G.

PUBLISHED PATENT APPLICATION

A Device and Method for Joint SOC and SOH Estimation of Lithium-Ion Batteries. Chinese invention patent application CN202210656043.X. Published in 2022.

SELECTED AWARDS

First-class National Award — 1st World University Underwater Robots Competition 2022

Provincial Second Prize — China Undergraduate Mathematical Contest in Modeling 2022

National Third Prize — Mathematical Modeling Competition in Three Northeastern Provinces 2022

National Bronze Award — 7th China International College Students' Internet+ Competition 2021

SELECTED MODELING PROJECTS

Composition Analysis and Identification of Ancient Glassware Sep 2022

Applied AdaBoost, K-nearest neighbors, logistic regression, and Pearson correlation analysis to composition-based classification and provenance questions. The project received a provincial second prize in the China Undergraduate Mathematical Contest in Modeling.

Optimal Utilization of Isolation Resources Apr–May 2022

Developed cellular-automaton and SEIkQRD models to estimate isolation-resource demand under different epidemic-control settings. The project received a national third prize in the Mathematical Modeling Competition in Three Northeastern Provinces.

METHODS AND TOOLS

Research methods	Deep learning · 3D U-Net · Vision Transformers · Radar semantic segmentation · SVM · Kalman filtering · Dempster–Shafer fusion · YOLOv8 · Model pruning · Edge deployment
Engineering tools	Python · MATLAB · ANSYS · SolidWorks · AutoCAD · Excel